Larkfield

US E-Commerce

CRISP-DM DESIGN DOCUMENT

Personalized Customer Response

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Course project — company and people renamed; based on a real company

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1. Data Architecture and Pipeline Design

1.1 Overview

The Larkfield Personalized Customer Response system is built on a modern cloud-based data architecture that ingests, transforms, and surfaces data across two internal systems. The architecture follows a medallion (Bronze / Silver / Gold) lakehouse pattern deployed on Databricks.

1.2 Data Flow Summary

Source systems feed a centralized data lake through rebuilt ETL pipelines. Transformed data flows into the feature store and model serving layer, then surfaces to the CX team dashboard and the customer response system.

Layer	Component	Description
Source	CX and customer platforms CX System, Customer Data System	Raw data on customer tickets, and all data on customers.
Ingestion (Bronze)	Databricks Autoloader + Databricks Delta Lake	Raw data landed unmodified; schema-validated at entry; full audit trail retained
Transformation (Silver)	Databricks Delta Live Tables + dbt	Cleaned, deduplicated, standardized; referential integrity enforced; PII masked
Feature Store (Gold)	Databricks Feature Store	Pre-computed model features at customer granularity; versioned; lineage tracked
Model Serving	Databricks Model Serving + MLflow Registry	Real-time and batch inference; version-controlled model registry
Presentation	Databricks SQL Dashboards (Lakeview) + Databricks Model Serving REST API	CX Team-facing UI; Databricks REST API Gateway called directly, and Customer Response Systems

1.3 Integration Points

- CX System → Bronze Schema (nightly batch extract)
- Customer Data System → Bronze (weekly full refresh)
- DeltaLake Endpoint → API Gateway → Databricks Unity Catalog (Databricks REST)

1.4 Data Lineage and Governance

Databricks Unity Catalog will serve as the central metadata repository. All datasets are registered with owner, classification, update cadence, and data quality scores. Databricks Unity Catalog enforces column-level access controls, ensuring that PII-bearing demographic fields are accessible only to authorized roles.

2. Model Architecture Specification

2.1 Problem Framing

The customer response task is decomposed into two tightly coupled sub-problems: (1) retrieving the most relevant past cases and customer context for each incoming ticket, and (2) generating a personalized response, or handing the ticket to a human agent when the system cannot resolve it.

2.2 Modeling Approach

Component	Chosen Approach	Rationale
RAG Layer	BGE-M3 Embedding + FAISS ANN Search + lightweight GPT-4o LLM Re-ranker	The embedder can chunk full customer records or information from each table. FAISS handles non-standard semantics. RAG output is ordered by relevance to improve the response
Personalized Response	Databricks Gemini Model Serving + Custom Prompt	LLMs can write personalized responses well given good context on the customer and instructions in the prompt
Bias Detection	LLM-as-a-Judge	Ensures responses do not drift over time.
Explainability	LIME (Local Interpretable Model-Agnostic Explanations)	Explains LLM response behavior using a simpler interpretable mode.

2.3 Feature Engineering

- Churn Rate: Percentage of Customers that do not purchase or keep their subscription over a window of time
- RFM: Recency, Frequency, and Monetary Composite feature based on behavior of customer
- Customer Lifetime Value: Estimated total discounted net revenue over a future horizon.

2.4 Target Variable

Customer Retention (Inverse of Churn Rate). Evaluation metric: Percentage of Customers evaluated to be saved by an LLM.

2.5 Model Registry and Versioning

All trained models are registered in Databricks MLflow Model Registry with training dataset version, hyperparameters, and evaluation metrics. Promotion from Staging to Production requires documented approval from the Data Science Lead and Business Owner.

3. Infrastructure and Technology Stack

3.1 Cloud Platform

All infrastructure is deployed on Databricks, selected following a vendor security assessment led by Alan Brooks (Security Specialist) and Elena Novak (Enterprise Architect). Databricks satisfies Larkfield's encryption, audit trail, and compliance requirements.

Category	Technology / Service	Purpose
Compute	Databricks Model Serving (training + endpoints)	Model training, hosting, and batch transform
Storage	Delta Lake on Amazon S3 (Bronze / Silver / Gold zones)	Data lake storage for all pipeline layers
ETL / Orchestration	Databricks Delta Live	Data transformation, pipeline scheduling, dependency management
Feature Store	Databricks Feature Store	Versioned feature storage and serving for training and inference
Database	Databricks SQL	Operational metadata, audit logs, response output cache
API Layer	Databricks Model Serving REST API	REST API endpoints for downstream system integration
Dashboard	Databricks SQL Dashboards	CX team dashboard
CI/CD	Databricks Asset Bundles + GitHub Actions	Automated model and application deployment pipeline
Monitoring	Databricks Lakehouse Monitoring + System Tables	Infrastructure alerts, model drift detection
Security	Databricks Unity Catalog + Cloud KMS + Databricks Secrets	Access control, encryption, secrets management
Development	Databricks PySpark Notebooks + Databricks IDE	Data science development environment
ML Libraries	MLflow, databricks-vectorsearch, pyspark.sql, sklearn	Core modeling and analysis stack

3.2 Environment Tiers

- Development: Databricks PySpark Notebooks (individual notebooks); ephemeral Delta Tables
- Staging: DLT Pipeline; Unity Catalog staging catalog Databricks Workflow test job + Staging serving endpoint for testing
- Production: HA-configured Databricks Model Serving + Auto-scaled endpoints; Delta Lake + Unity Catalog with S3 versioning and object lock; Delta Transaction Log

4. Data Quality and Validation Framework

4.1 Framework Overview

Data quality is enforced at each pipeline layer using DLT Expectations for automated rule-based validation, with results published to a central quality dashboard. Each dataset has a defined Data Owner accountable for resolution of quality failures.

4.2 Validation Rules by Dataset

Dataset	Key Validation Rules	Failure Action
Purchases	No nulls on customer_id, sku_id, date, num_purchased; date within expected range; no duplicate (customer id, date) records	Block pipeline; alert Priya Nair; quarantine bad records
CX Tickets	All tickets logged, customer_id maintained, data included. PII masked	Alert Priya Nair; Notify if PII is breached.
Customer Returns	Unified returns data across client systems.	Alert Tom Reyes; use last-known-good record until resolved
Campaigns	Date, platform used, and customer id intact	Block ingestion; alert Alan Brooks; legal notification if PII breach detected
Subscriptions	Subscription, e-commerce brand its from, and customer id	Alert Priya Nair

4.3 Data Cleansing Procedures

- Customer Returns: Unified across internal database and help desk database.
- ETL: Idempotent pipeline design ensures reruns do not duplicate records; all transforms logged to audit table. Scalable ETL Pipeline needed complete with CI/CD integration capabilities. Uses LLM-as-a-warning system for downtime detection and immediate repair analysis.

5. Security and Privacy Controls

5.1 Regulatory Compliance

The solution must comply with the PCI-DSS Compliance and applicable federal and international data privacy regulations. Formal legal review and sign-off by Alan Brooks and the Larkfield Legal Team is a deployment gate preceding any production launch.

5.2 Data Protection Controls

Control	Implementation	Owner
Encryption at Rest	AWS KMS-managed AES-256 encryption on all Delta Lake S3 buckets	Alan Brooks / IT
Encryption in Transit	TLS 1.3 enforced on all Databricks REST API endpoints and inter-service communication. Cluster-to-Cluster encryption	Alan Brooks / IT
PII Masking	Customer demographic PII masked via Databricks Unity Catalog AI-as-a-PII Masker at Bronze-to-Silver boundary before any model use.	Alan Brooks
Access Control	Databricks Unity Catalog RBAC/ABAC roles with least-privilege; IAM retained on the infra layer; system table access; SSO+MFA required for all engineer access via IDP	Alan Brooks / IT
Audit Logging	CloudTrail enabled on all AWS services; all data access events logged to immutable S3 audit bucket	Alan Brooks
Bias Audit	Fairlearn demographic parity checks run at each model training cycle; results reviewed before promotion. Databricks MLflow logged metrics + Databricks Lakehouse Monitoring	Priya Nair / Legal
Secrets Management	Database credentials and API keys stored in Databricks Secrets API; rotated quarterly. Scope-based access ensures principle of least privilege.	IT
Vendor Assessment	Databricks provider security assessment completed prior to production data upload; Databricks SOC 2 Type II + ISO 27001 required	Alan Brooks / Elena Novak
Incident Response	Defined IR playbook; 24-hour breach notification protocol; Databricks Alerts alarms trigger PagerDuty alerts.	Alan Brooks / IT
Consent Mechanisms	Consent status tracked on each customer record; records with no consent excluded from CRM-sourced features. Enforced with Databricks Unity Catalog row filters	Legal / Alan Brooks

5.3 Supplier / Vendor Contract Review

The Legal Team will review all supplier and vendor data sharing agreements to confirm that planned data usage is consistent with contractual terms. This review is targeted for completion within Phase 1 in parallel with ETL pipeline redesign.

6. Deployment Strategy and Environment Setup

6.1 CI/CD Pipeline

Model and application deployments are automated through GitHub Actions+Databricks with the following stages: Source (GitHub merge to main) → Build (GitHub Actions: unit tests, lint, asset bundle validation) → Staging Deploy → Integration Tests → Manual Approval Gate → Production Deploy.

6.2 Environment Configuration

Environment	Purpose	Data Scope	Access
Development	Feature development; exploratory analysis	Anonymized 10% data sample	Data Science Team
Staging	Integration testing; pre-production validation	Full dataset (masked PII)	Data Science + IT + QA
Production	Live customer response	Full production data	Controlled; MFA required

6.3 Rollback Plan

- Model rollback: Previous model version remains registered in Databricks MLflow Model Registry. Traffic can be shifted back within 15 minutes via serving endpoint update.
- Application rollback: Previous Databricks Asset Bundle retained in version control; rollback re-deploys prior approved build within 20 minutes.
- Data rollback: S3 object versioning enables Databricks Delta time travel.
- Trigger criteria: Transfers to real CX agents > 10% on live data; dashboard error rate > 1%; CX system integration failure affecting > 2 client brands.

7. Monitoring and Maintenance Plan

7.1 Performance Monitoring

Metric	Target	Monitoring Tool	Alert Threshold	Owner
Model Transfer Rate (live)	< 5%	Databricks Lakehouse Model Monitor.	Transfer Rates > 10% for 2 consecutive weeks	Priya Nair
Data Drift (feature distributions)	< 10% PSI vs. training baseline	Databricks Alert with built-in PSI monitoring	PSI > 0.2 on any key feature	Priya Nair
Pipeline Latency (nightly ETL)	< 2 hours	Databricks Workflow	Duration > 3 hours	IT
API Response Time (p99)	< 500ms	Databricks serving endpoint	> 1000ms sustained	IT
Dashboard Uptime	99.5%	Databricks Lakehouse Monitoring	Consecutive failures > 5 minutes	IT
Response Coverage	> 95% of eligible tickets	Custom Databricks Delta Table Metric + Databricks Alert SQL query, layered with DLT Expectation	< 90% coverage	Priya Nair

7.2 Model Retraining Schedule

- Scheduled retraining: Quarterly full retrain on rolling 104-week history
- Triggered retraining: Initiated automatically when data drift PSI > 0.2 or Failure Rate exceeds alert threshold for two consecutive monitoring cycles
- Retrain approval: Automatic promotion to staging; manual sign-off from Data Science Lead required for production promotion

7.3 Maintenance Responsibilities

Activity	Frequency	Owner
ETL pipeline health check	Daily (automated)	IT / Databricks Workflows alerts
Data quality report review	Weekly	Priya Nair
Model performance review	Bi-weekly	Priya Nair
Bias and fairness audit	Quarterly	Priya Nair + Legal
Security audit and penetration test	Semi-annually	Alan Brooks + External Vendor
Dependency updates (libraries, OS)	Monthly	IT
Dashboard UX feedback review	Quarterly	Tom Reyes

8. Testing and Validation Approach

8.1 Testing Levels

Test Level	Scope	Tools	Pass Criteria
Unit Tests	Individual ETL transforms, feature engineering functions, API handlers	pytest, Databricks dbutils MagicMock	100% pass; > 85% code coverage
Integration Tests	End-to-end pipeline from source ingestion to API response; CX System and Customer Data System API mocks	pytest + Databricks Workflows	All integration scenarios pass in staging environment
Model Validation	Out-of-sample held-out test set (most recent 13 weeks excluded from training); walk-forward cross-validation	scikit-learn, Databricks Mosaic AI Vector Search + Databricks ai_query LLM-as-a-Judge evaluation	Predicted failure < 10% on held-out test set across all client brands
Bias / Fairness Tests	Demographic parity checks on response quality across customer segments	Fairlearn, MLflow metrics logging for parity ratios	Demographic parity ratio within 0.80–1.25 across all segments
Performance / Load Tests	API endpoint load simulation at 2x expected peak concurrent users	Locust, Databricks Model Serving Endpoint	p99 latency < 500ms at 50 concurrent users
User Acceptance Testing	Customer and client pilot using real CX ticket evaluation	Structured feedback survey	Average satisfaction score >= 3.5 / 5; < 20% negative feedback on usability

8.2 Acceptance Criteria for Production Promotion

- Predicted response failure < 10% verified on held-out test set
- All unit and integration tests passing in staging
- Legal sign-off on model logic and data usage obtained (Alan Brooks + Legal Team)
- Bias audit passed with Fairlearn demographic parity within specified bounds
- Load test passing at 2x expected peak load
- UAT satisfaction score >= 3.5/5 from pilot clients and the CX team

9. API and Interface Specifications

9.1 API Design Principles

All external interfaces use RESTful JSON APIs over HTTPS (TLS 1.3). Authentication uses Databricks PATs + Databricks OAuth M2M API Gateway authorizers with per-consumer API keys. Rate limiting is set at 100 requests/minute per consumer. API versioning follows the /v1/ path prefix convention.

9.2 Key Endpoints

Endpoint	Method	Consumer	Description	Response Format
/v1/response/{customer_id}	GET	Dashboard; APIM	Returns customer responses.	JSON: response text, retrieved case IDs, handoff flag, LIME explanation
/v1/data-quality/status	GET	Internal monitoring	Returns current data quality scores and pipeline health by dataset	JSON: dataset quality summary
/v1/model/metrics	GET	Internal monitoring	Returns current live model performance metrics (Transfer Rate, coverage, drift scores)	JSON: model performance summary

9.3 Dashboard Interface

The CX team-facing responses dashboard is built on Databricks SQL Dashboards. It surfaces aggregated insights on customers with LIME explanations for responses. All interactions are logged for audit and feedback analysis.

10. Scalability and Performance Considerations

10.1 Current Scale Baseline

The initial deployment serves high value customers and influencers. The primary computational workload is the nightly batch inference run and quarterly model retraining.

10.2 Scaling Strategy

Dimension	Current Requirement	Scaling Approach
High Value Customers	20,000 customers	Databricks AutoLoader complete ETL AutoScaling; linear cost scaling — no re-architecture needed up to ~100,000 customers
API Throughput	~50 concurrent dashboard users during peak	Databricks Model Serving endpoint auto-scaling configured
Data Volume	~3 years of historical customer and ticket data	Databricks Unity Catalog + Delta Lake with liquid clustering. Uses Databricks auto-scaling
Training Frequency	Quarterly full retrain; triggered retrain as needed	Databricks ML Runtime managed training; spot instances used for cost reduction (~60% savings vs. on-demand)

10.3 Performance Benchmarks

- Nightly batch inference: < 30 minutes for all active customers
- Dashboard page load (customer response view): < 2 seconds
- API p50 latency: < 150ms; p99 latency: < 500ms
- Quarterly model retrain: < 4 hours

11. Risk Mitigation and Contingency Plans

Risk	Likelihood	Impact	Mitigation	Contingency Owner
ETL pipeline delays block model development	High	High	Phase 1 critical path; dedicated sprint; Elena Novak + Priya Nair co-own pipeline rebuild; weekly progress check-ins with Tom Reyes	Elena Novak
Legal review delays deployment	Medium	High	2–4 week legal buffer embedded in timeline; legal review initiated in Phase 2 in parallel with model development	Alan Brooks + Legal
Failure Rate fails to achieve < 10% target	Medium	High	LLM-as-a-Judge evaluation; switch to Claude Opus 4.7 if the current model underperforms; prompt and retrieval iteration planned	Priya Nair
Cloud provider security clearance delayed	Medium	High	Vendor assessment initiated in Phase 1; fallback to alternative approved vendor if primary fails; on-premise staging as bridge	Alan Brooks + Elena Novak
Client / CX team low adoption	Medium	Medium	LIME plain-language explanations in dashboard; structured 15-hour training program; pilot phase feedback loop; change management plan	Tom Reyes
Data quality failures (PII breach)	Low	Critical	PII masking at ingestion boundary; Unity Catalog access controls; CloudTrail audit; incident response playbook; legal notification within 24 hours	Alan Brooks
Budget overrun	Low	Medium	Finance approval of detailed budget before procurement; Spot instances for training cost reduction; monthly cost review with Tom Reyes	Tom Reyes + Finance
Algorithmic bias in responses	Low	High	Fairlearn bias checks at every model training cycle; demographic parity monitoring post-deployment; quarterly bias audit with legal	Priya Nair + Legal
IT capacity constraints delay integrations	Medium	Medium	Formal IT scheduling and capacity commitment from Elena Novak; integration work scoped into Phase 4 with 2-week buffer	Elena Novak + IT

12. Pilot Phase Implementation Plan

12.1 Pilot Scope

The pilot phase is executed during Weeks 13–16 of the project (Phase 4) following successful completion of model validation and integration testing. It is designed to validate personalized responses under real operational conditions before full-enterprise rollout.

Dimension	Pilot Definition
Target Customers	Top 20,000 customer by CLV.
Feature Set	Full response system including RAG retrieval, customer responses, and LIME explanations
Duration	4 weeks of live personalized customer responses (Weeks 13–16
User Groups	Customers that are high value
Rollback Trigger	Failure Rate > 15% on live pilot data; customer satisfaction < 2.5/5; integration failure affecting > 1 client brand for > 24 hours
Advancement Criteria	Failure Rate < 15% maintained during pilot; client satisfaction >= 3.5/5; integration stable; compliance audit passed; Legal sign-off received

12.2 Pilot Timeline (Weeks 13–16)

Week	Activities
Week 13	Model integration into CX and Customer Data systems completed; dashboard deployed to staging; CX team training delivered (15 hours); pilot client briefings completed
Week 14	Pilot go-live: personalized responses begin for the top 20,000 customers; CX team begins reviewing dashboard; daily data quality monitoring; compliance audit check initiated (Alan Brooks)
Week 15	Mid-pilot review: Failure Rate and data quality assessment; structured client and CX team feedback collection (mid-point survey); any critical issues escalated for immediate remediation
Week 16	Pilot conclusion: final performance metrics compiled; post-pilot satisfaction survey administered; compliance audit finalized; pilot outcomes report prepared for executive team

12.3 Pilot Rollback Procedure

- Step 1: Incident declared by Data Science Lead or IT on-call (within 2 hours of trigger threshold breach)
- Step 2: Databricks Serving endpoint traffic shifted to human CX agent fallback within 15 minutes
- Step 3: CX team notified via dashboard banner to revert to manual ticket handling
- Step 4: Root cause analysis initiated; remediation target < 5 business days
- Step 5: Re-entry into pilot requires sign-off from Priya Nair, Tom Reyes, and Alan Brooks

13. Pilot Phase Business Value Estimation

13.1 Baseline Measurements

The following baseline metrics will be established during Phase 1 (Weeks 1–3) using pre-project historical data from all client brands. Pilot-phase performance will be measured against these baselines to determine actual vs. expected outcomes.

KPI	Baseline Measurement Method	Baseline Period
Project Customer Retention Rate%	Average customer retention rate across client brands	26 weeks prior to Phase 1
Failure Rate	Percent of customer issues that are not resolved over email and text and need a real CX person	13 weeks prior to Phase 1
CLV	Customer Lifetime Value.	26 weeks prior to Phase 1

13.2 Projected Pilot Value

The pilot covers 20,000 customers over 4 weeks. Pilot cost is estimated from the full-rollout response cost. Retention value is not projected until Phase 1 establishes a baseline.

Value Driver	Full-Project Annual Estimate	Pilot (20,000, 4 weeks)	Methodology
Response system cost	$50,000 annually across all customer service	$10,000	Applied to 20,000 customers. Response System uses a linear scaling meaning 20,000 customers costs $10,000 at a cost of $0.50 per response
TOTAL PILOT ESTIMATED COST	—	$10,000	Cost only. Retention value is measured against the baseline during the pilot.

13.3 Measurement Methodology

- Customer Retention: Compare customer retention % for pilot customers during pilot weeks vs. same-period prior-year baseline, adjusted for seasonal index
- Failure Rate : Compute weekly on live customer response failure vs. AI evaluation ; track against < 10% target
- Satisfaction: Post-pilot survey (1–5 Likert scale) administered to all pilot clients

14. Production Phase Implementation Plan

14.1 Rollout Strategy

Following a successful pilot with all advancement criteria met, full production rollout is executed during Weeks 17–24 (Phase 5). A phased customer-cohort activation approach is used to manage operational risk and allow for course correction.

Week	Customers Active	Key Activities
17–18	40,000 customers total	Executive pilot outcomes presentation; stakeholder approval to proceed; onboard second cohort; CX team training; dashboard access provisioned
19–20	80,000 customers total	Onboard third cohort; continued monitoring; mid-rollout performance review with Tom Reyes; any integration issues from new clients remediated
21–22	100,000 customers (all)	Final customers involved; all clients and CX team trained and using dashboard
23–24	100,000 customers (stabilization)	Optimization iterations based on full-rollout feedback; final KPI performance measurement; post-implementation review; executive report delivered to Dana R.

14.2 User Onboarding and Change Management

- CX team training (15 hours) delivered in cohorts aligned to the cohort activation schedule
- Quick-reference dashboard guide distributed in print and digital formats to all clients
- Weekly all-hands update email from Tom Reyes during rollout period
- Dedicated Slack channel for client questions and model feedback during rollout weeks
- Executive dashboard KPI summary delivered bi-weekly to Dana R. (CFO)

14.3 Production Infrastructure Scaling

- Databricks serving endpoint auto-scaling enabled: minimum 1 instance, maximum 4 instances (m5.xlarge), scale-out at 70% CPU
- Databricks nightly ETL pipeline configured for all customers
- Delta Lake time travel retained for 30 days
- Databricks REST API Gateway throttling: 500 req/s burst; 100 req/s sustained per consumer
- S3 object versioning and replication to secondary Databricks region (us-east-1 primary; us-east-2 replica)

14.4 Data Migration and Cutover

No destructive data migration is required. The new pipeline reads from existing source systems. At production cutover, Databricks Workflow schedules are activated for all customers and the dashboard URL is distributed to the CX team and clients. The existing playbook chatbot and human-agent process is maintained in parallel for 4 weeks post-cutover as a fallback.

15. Post-Production Monitoring Plan

15.1 Automated Monitoring Architecture

Post-production monitoring is implemented through a three-tier observability stack: (1) infrastructure and pipeline health via Databricks system tables, (2) model performance and data drift via Databricks Lakehouse Model Monitor, and (3) business KPI tracking via a dedicated executive Databricks SQL dashboard.

15.2 Monitoring Dashboard and Alerts

Monitoring Domain	Key Metrics	Alert Channel	Response SLA
Infrastructure Health	ETL pipeline duration; API error rate; endpoint CPU/memory; S3 ingestion success rate	Databricks Alerts → PagerDuty → IT On-Call	15 minutes (P1 outage); 4 hours (P2 degraded)
Model Performance	Live Failure Rate by client brand; feature drift (PSI); response coverage (%); prediction confidence distribution	Databricks Alerts → Email → Priya Nair	Review within 24 hours; retrain decision within 48 hours
Data Quality	DLT Expectations results; null rate by dataset; schema drift; pipeline freshness	Databricks Alerts → Email → Priya Nair + Data Owner	Dataset owner notified within 1 hour; resolution SLA: 4 hours
Business KPIs	Failure Rate trend; customer response acceptance rate	Bi-weekly report → Tom Reyes + Dana R.	Quarterly formal review; ad-hoc alerts on > 5% KPI deviation
Security and Compliance	Unauthorized access attempts; unusual API call patterns; PII access audit; CloudTrail anomalies	Databricks Alerts → PagerDuty → Alan Brooks	Immediate (P1 security event); daily review (audit log anomalies)

15.3 Model Drift and Retraining Protocol

- Databricks Lakehouse Model Monitor runs nightly baseline comparison against training-data feature distributions
- PSI (Population Stability Index) > 0.2 on any primary feature triggers automatic retraining job
- Failure Rate exceeding 18% for two consecutive monitoring cycles triggers retraining and Data Science Lead notification
- Quarterly scheduled retraining regardless of drift signals to incorporate latest customer and ticket data
- All retrained models evaluated on held-out test set before any production promotion; manual approval required

15.4 Continuous Improvement Cadence

Activity	Cadence	Owner	Output
Model performance review	Bi-weekly	Priya Nair	Performance memo; retrain decision
Client and CX team feedback analysis	Monthly	Tom Reyes	Dashboard UX improvement backlog
Bias and fairness audit	Quarterly	Priya Nair + Legal	Fairness audit report; legal certification
Security and penetration test	Semi-annually	Alan Brooks + External Vendor	Security assessment report
Executive KPI review	Quarterly	Tom Reyes + Dana R.	Strategic alignment assessment; project extension decision
Full post-implementation review	12 months post-launch	All stakeholders	Lessons learned report; roadmap for Phase 2 expansion

16. Document Sign-Off

This CRISP-DM Design Document has been reviewed and is approved for implementation. By signing below, the undersigned confirm that they have reviewed the design, agree with the technical and business approach described, and authorize the project to proceed to the execution phases outlined herein.

Signatures confirm approval of:

- Data architecture, pipeline design, and technology stack
- Model architecture, feature engineering approach, and evaluation criteria
- Security, privacy, and compliance controls
- Pilot and production deployment strategy and rollout plan
- Post-production monitoring and maintenance commitments

Role	Name & Title	Signature & Date
Executive Sponsor	Dana R., CFO	_____
Tactical Manager / Project Lead	Tom Reyes, Tactical Manager	_____
Data Science Lead	Priya Nair, Business / Data Specialist	_____
Security & Compliance	Alan Brooks, Security Specialist	_____
Enterprise Architect	Elena Novak, Enterprise Architect	_____
Legal Representative	TBD — Legal Team	_____
Finance Representative	TBD — Finance	_____

Notes / Conditions of Approval:

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— End of Design Document —

Course project — company and people renamed; based on a real company